Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Artificial Intelligence Application Development and Jira

Practical – IV (TASK)

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Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

CODE:

```
# Probability of knowing Python
P_Python = 0.20

# Probability of knowing Java
P_Java = 0.80

# Probability of getting placed if student knows Python
P_Placed_Python = 0.90

# Probability of getting placed if student knows Java
P_Placed_Java = 0.40

# Bayes' Theorem
P_Python_Placed = (P_Placed_Python * P_Python) / (
    (P_Placed_Python * P_Python) +
    (P_Placed_Java * P_Java)
)

# Print answers in percentage
print("Probability that the student knows Python:", round(P_Python * 100, 2), "%")
print("Probability that the student knows Java:", round(P_Java * 100, 2), "%")
print("Probability that a placed student knows Python:", round(P_Python_Placed * 100, 2),
"%")
```

OUTPUT:

```
...
Probability that the student knows Python: 20.0 %

Probability that the student knows Java: 80.0 %

Probability that a placed student knows Python: 36.0 %
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

CODE:

```
# Probability that the taxi is Blue
P_Blue = 0.15

# Probability that the taxi is Green
P_Green = 0.85

# Probability that the witness correctly identifies Blue
P_SaysBlue_Blue = 0.80

# Probability that the witness says Blue when the taxi is actually Green
P_SaysBlue_Green = 0.20   # (1 - 0.80)

# Bayes' Theorem
P_Blue_SaysBlue = (P_SaysBlue_Blue * P_Blue) / (
  (P_SaysBlue_Blue * P_Blue) +
  (P_SaysBlue_Green * P_Green)
)

# Display the result
print("Probability that the taxi was actually Blue:",
    round(P_Blue_SaysBlue * 100, 2), "%")
```

OUTPUT:

```
...  Probability that the taxi was actually Blue: 41.38 %
```